

Comparing Vaccination and *Wolbachia* Releases for Dengue Control in Indonesia

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Contributions to code files

process-demography.R was provided by Emilie Finch. Early versions of the code files 00_load-packages.R, config.yaml, env-model.R, run-model.R and utils.R were provided by Emilie Finch. These files have been heavily modified. run-model.R has been split into various files, including 02_run_batch.R, equilibrate_sim.R and scenario_sim.R. ChatGPT (OpenAI, 2026) and Claude.ai (Anthropic, 2026) were used for code generation and debugging. I checked and modified all AI generated code. Descriptions of the purpose of each file can be found in the “Code files” section of the supplementary information.

The code can be found at: <https://github.com/anonymous1662/denv-indonesia>.

Abstract

The public health burden due to dengue is increasing. A recently licensed vaccine, TAK-003, and a novel *Wolbachia* technology offer potential means to limit the burden of dengue. However, the differential impacts of these intervention programmes have not been well explored and the optimal manner regarding if and how to combine them is unclear. Using a stochastic transmission model with an S-I-C-R-V framework, we simulated future dengue burden in Indonesia. We developed criteria to identify priority areas for *Wolbachia* deployment and produced a map of operationally feasible municipalities for *Wolbachia* release. We estimate that a wMel release campaign targeting 100 municipalities could avert 30 – 32% of cases over the next 15 years, while a national vaccination campaign primarily targeting 10-year-olds could avert 5 – 6% of cases in the total population and 63% of cases in the vaccinated cohort. We find that a strategy combining both interventions would likely be optimal and could avert higher proportions of cases with 34 – 36% averted in the total population and 74 – 75% averted in the vaccinated cohort. We find that interventions could drive up the mean age of dengue cases beyond increases due to ageing in the population. Longer-term modelling is needed to assess how the intervention benefits change over multiple generations.

Introduction

The global burden of dengue is rising, with over half the current global population predicted to be at-risk (Lim et al., 2025). Despite this, the development of effective interventions for dengue control has been challenging. Traditional vector control methods based around the suppression of mosquito populations have been largely ineffective in the long-term (Esu et al., 2010). Vaccine development efforts have been ongoing for over 60 years (Bellanti et al., 1966) but have been complicated by interactions between serotypes (Goncalvez et al., 2007; Izmirly et al., 2020). However, second generation vaccines and a novel biocontrol method involving release of *Wolbachia* have recently shown more promise.

TAK-003 is the only licensed vaccine currently commercially available (Wilder-Smith et al., 2025). A phase III clinical trial involving approximately 21,000 healthy 4–16-year-olds found TAK-003 had a cumulative efficacy after 3 years of 62.0% against virologically confirmed dengue (VCD) and 83.6% against hospitalised VCD (Rivera et al., 2022). However, efficacy varied by serostatus and serotype, with lower values in seronegative recipients. Although results after 4.5 years indicate a lack of efficacy against DENV-3 in seronegative recipients and inconclusive results for DENV-4, they did not indicate increased disease severity (Tricou et al., 2024). However, there are concerns that these estimates may represent a weak signal of vaccine-associated disease enhancement, masked by low case numbers (Daniels et al., 2025). Nonetheless, WHO recommends that serostatus pre-screening is not required with vaccination (World Health Organization, 2024).

Releases of *Wolbachia*-carrying *Aedes aegypti* mosquitoes have been trialled as a novel dengue control strategy. *Wolbachia* are intracellular maternally-inherited bacteria (Utarini et al., 2021) which confer a reproductive advantage to their *Aedes aegypti* hosts (Ross et al., 2022), facilitating introgression into wild populations. Establishment of *Wolbachia* in mosquito populations reduces the transmission intensity of dengue (Ferguson et al., 2015). Introgression of wMel into intervention clusters showed 77.1% efficacy for the prevention of symptomatic VCD in a trial conducted in Yogyakarta, Indonesia (Utarini et al., 2021). However, local introgression is not always successful and there is evidence that efficacy is temperature-dependent (Ross et al., 2017).

It is currently unclear how best to combine these interventions to maximise public health impact. Indonesia has a large dengue burden, with an estimated 7.8 million symptomatic cases in 2015 (O'Reilly et al., 2019). In this study, we quantify potential impacts of routine vaccination with TAK-003 and wMel releases in Indonesia. Additionally, we produce maps of operationally feasible target areas for *Wolbachia* release considering population density and force of infection (FOI) estimates.

We used a stochastic S-I-C-R-V transmission model framework, informed by subnational FOI estimates, demographic projections and intervention efficacy estimates, to simulate dengue dynamics

within 500 municipalities. We assess the cases and hospitalisations averted to evaluate the public health impact of different intervention scenarios. We also assess the impact of each intervention on the mean age of cases.

Results

Model set-up and municipality selection for *w*Mel release

We projected infections, symptomatic cases, and hospitalisations across 500 municipalities using a stochastic transmission model (Methods). We ran 500 stochastic simulations for 15 years, starting from 2025, after a 150-year equilibration period (Figure S1). We compared a baseline scenario with no interventions to: a national vaccination scenario with 60% coverage, assuming waning over 10 years; a *w*Mel scenario with releases in 100 municipalities; a combined national vaccination and *w*Mel release scenario. We assumed that vaccination protects against symptomatic disease and hospitalisation but not infection, with an efficacy profile based on TAK-003 (Table S1). We assumed that vaccine efficacy varies by serostatus, with higher efficacy in seropositive recipients. Vaccination was modelled as a national 3 year catch up campaign targeting 10–16-year-olds, followed by national routine vaccination of 10-year-olds. *w*Mel-carrying mosquitoes were assumed to introgress successfully into every targeted municipality and reduce R_0 by 50%. We selected 100 municipalities for *w*Mel roll-out based on population-weighted density and FOI estimates (Methods, Figure 1a, Figures S2 and S3), with selected municipalities shown in Figure 1b. This set of municipalities covers 29.3% of the national population (Figure S4).

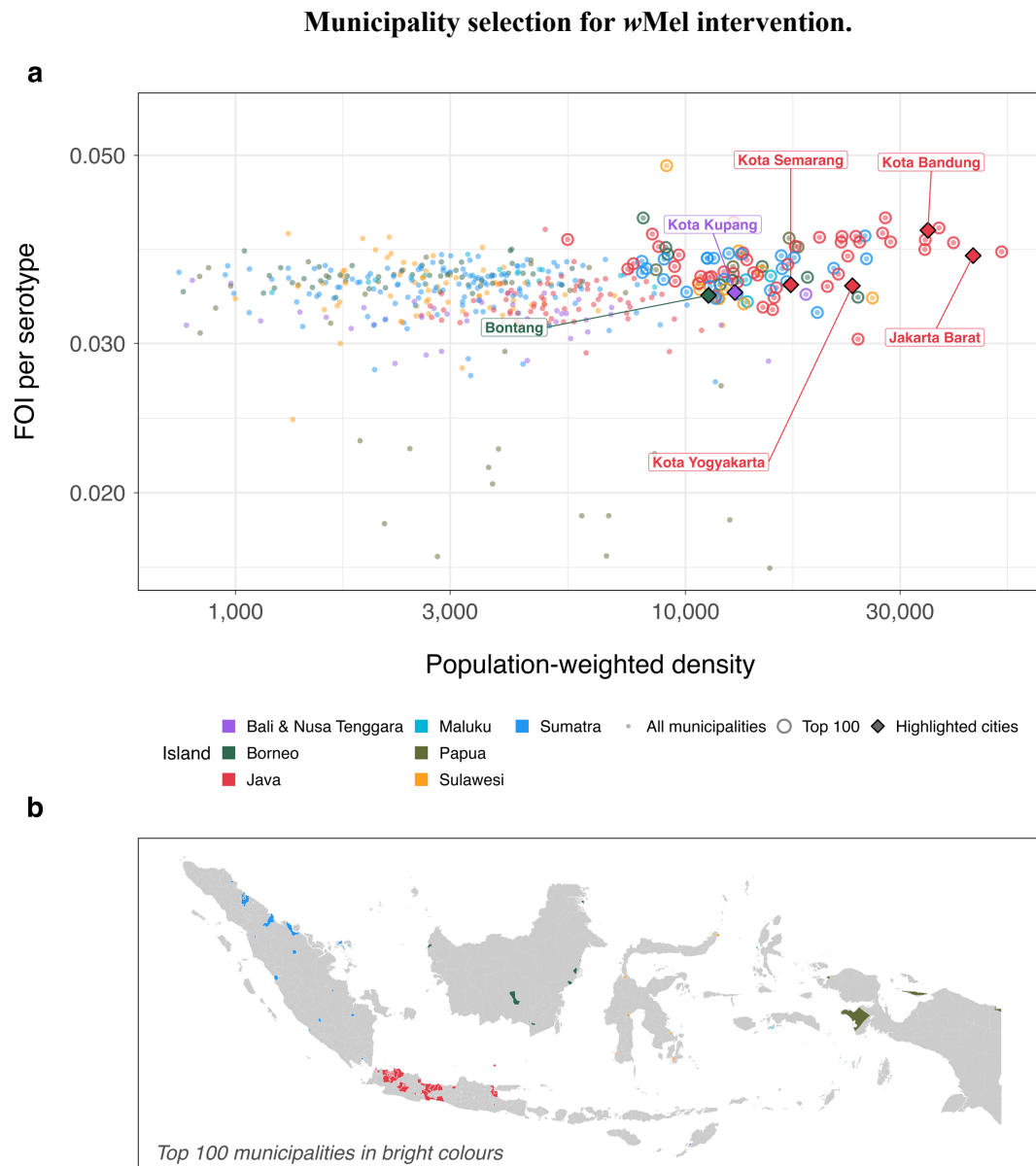


Figure 1 | (a) Scatter plot of municipalities in Indonesia by population-weighted density (on the x-axis, with a log 10 scale) and FOI per serotype (on the y-axis), calculated as total FOI / 4. The points are coloured by island and the top 100 ranking municipalities are highlighted (large circles). Additionally, municipalities where wMel release has been or is currently being trialled are highlighted (diamonds) (World Mosquito Program Ltd, 2025). **(b)** The locations of the 100 municipalities selected for wMel intervention in our simulations coloured by island.

National cumulative burden across 15 years

To assess the impact of interventions over the simulation period, we summarised the cumulative cases and hospitalisations each year by taking the median and 95% uncertainty intervals (UI) across the 500 simulations (Figure 2). We assessed both the population- and individual-level impacts. Population-level burden refers to the total burden across the entire population. Individual-level burden refers to the burden across vaccinated individuals only (or the equivalent population in scenarios without vaccination).

Across the 15-year projection period, there was a steady increase in dengue burden, reaching approximately 45.5 million cumulative symptomatic cases (Table 1) and over 4 million hospitalisations (Table S2) in the baseline scenario at the population-level. At the population-level, the cumulative burden was lowest in the combined strategy throughout the simulation, followed closely by wMel-only. The cumulative burden in the vaccination scenario was only slightly lower than the baseline. However, at the individual level, vaccination produced larger reductions in burden than wMel-only.

Cumulative burden at the population- and individual-level across 15 years

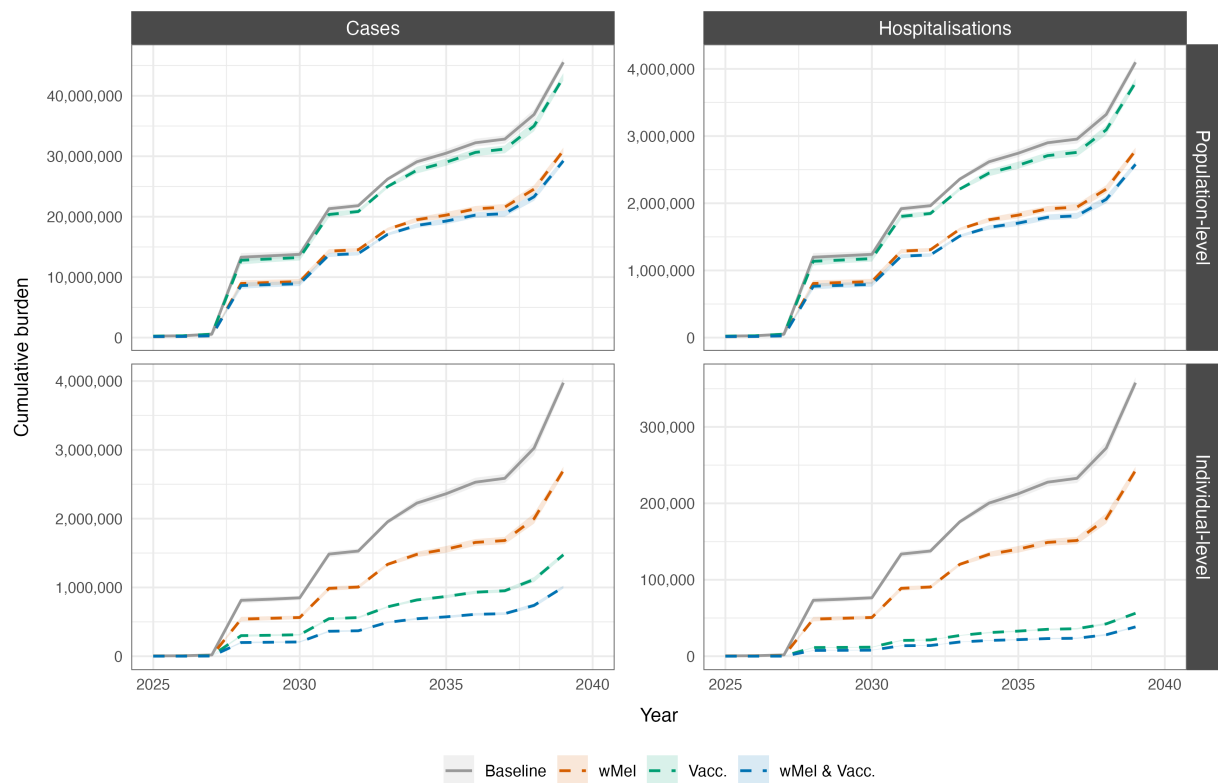


Figure 2 | The national median cumulative burden as cases (left) and hospitalisations (right) over time (15 years), comparing no intervention (grey), wMel only (orange), vaccination only (green) and wMel & vaccination (blue). 95% UI are shown. The top row shows the population-level impact and the bottom row shows the individual-level impact.

Population-level cumulative cases and hospitalisations by scenario

Scenario	Cases	Hospitalisations
Baseline	45,546,273 (44,961,246 – 46,210,141)	4,099,165 (4,046,512 – 4,158,913)
wMel	30,942,105 (30,446,157 – 31,417,860)	2,784,789 (2,740,154 – 2,827,607)
Vacc.	43,028,541 (42,506,896 – 43,735,939)	3,796,297 (3,750,398 – 3,858,233)
wMel & Vacc.	29,236,572 (28,803,549 – 29,705,648)	2,579,104 (2,541,266 – 2,620,411)

Table 1 | The population-level national median (95% UI) cumulative cases and hospitalisations over 500 stochastic simulations for the 15-year simulation period.

National public health impacts after 15 years

To quantify the impact of each intervention after 15 years, we calculated the proportional reduction in burden relative to the baseline scenario for each stochastic simulation, at both the population- and individual-level. We summarised the results by taking the median and 95% UI (Figure 3).

At the population-level, we found that the *w*Mel campaign targeting 100 municipalities would avert 31.4 % of cumulative cases (95% UI: 30.5 – 32.3), while the national vaccination program would avert 5.5% (95% UI: 5.0 – 5.9) and the combined intervention would avert 35.2% (95% UI: 34.4 – 36.0). Similar patterns were observed for hospitalisations.

In contrast, individual-level impacts were substantially larger for vaccination. We found that the *w*Mel campaign would avert 31.5% of cumulative cases (95% UI: 30.3 – 32.6), while the national vaccination campaign would avert 62.9% (95% UI: 62.7 – 63.1) and the combined intervention would avert 74.5% (95% UI: 74.1 – 74.9).

National public health impacts of wMel releases and vaccination over 15 years

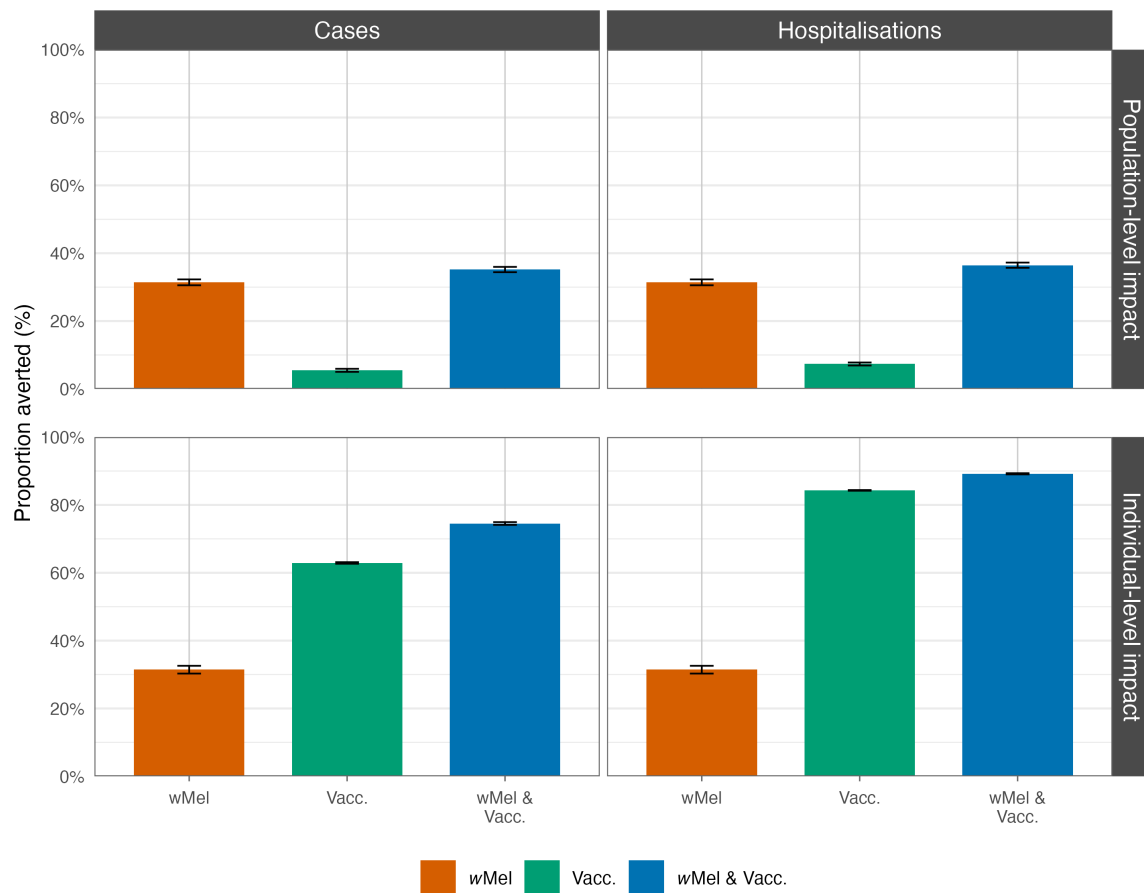


Figure 3 | The national proportion of cases and hospitalisations averted comparing wMel only (orange), vaccination only (green) and vaccination & wMel (blue). The bars show the median proportion averted for each intervention scenario compared to no intervention over 500 stochastic simulations. The error bar shows the 95% UI. Population-level impact (top row) refers to the proportion averted across the entire population, while individual-level impact (bottom row) refers to the proportion averted within the vaccinated or ‘would-have-been-vaccinated’ cohort.

Municipality-level impacts after 15 years

To assess spatial variation in intervention impact, we estimated the population-level median cases averted per 10,000 individuals after 15 years for each municipality (Figure 4).

Vaccination produced relatively uniform but modest reductions across municipalities, while *w*Mel averted large case numbers only in municipalities where releases were implemented. The combined strategy produced the largest overall reductions, with the highest impact areas reflecting spatial patterns of *w*Mel intervention on a background of smaller reductions from vaccination. Similar spatial patterns were observed for hospitalisations (Figure S5).

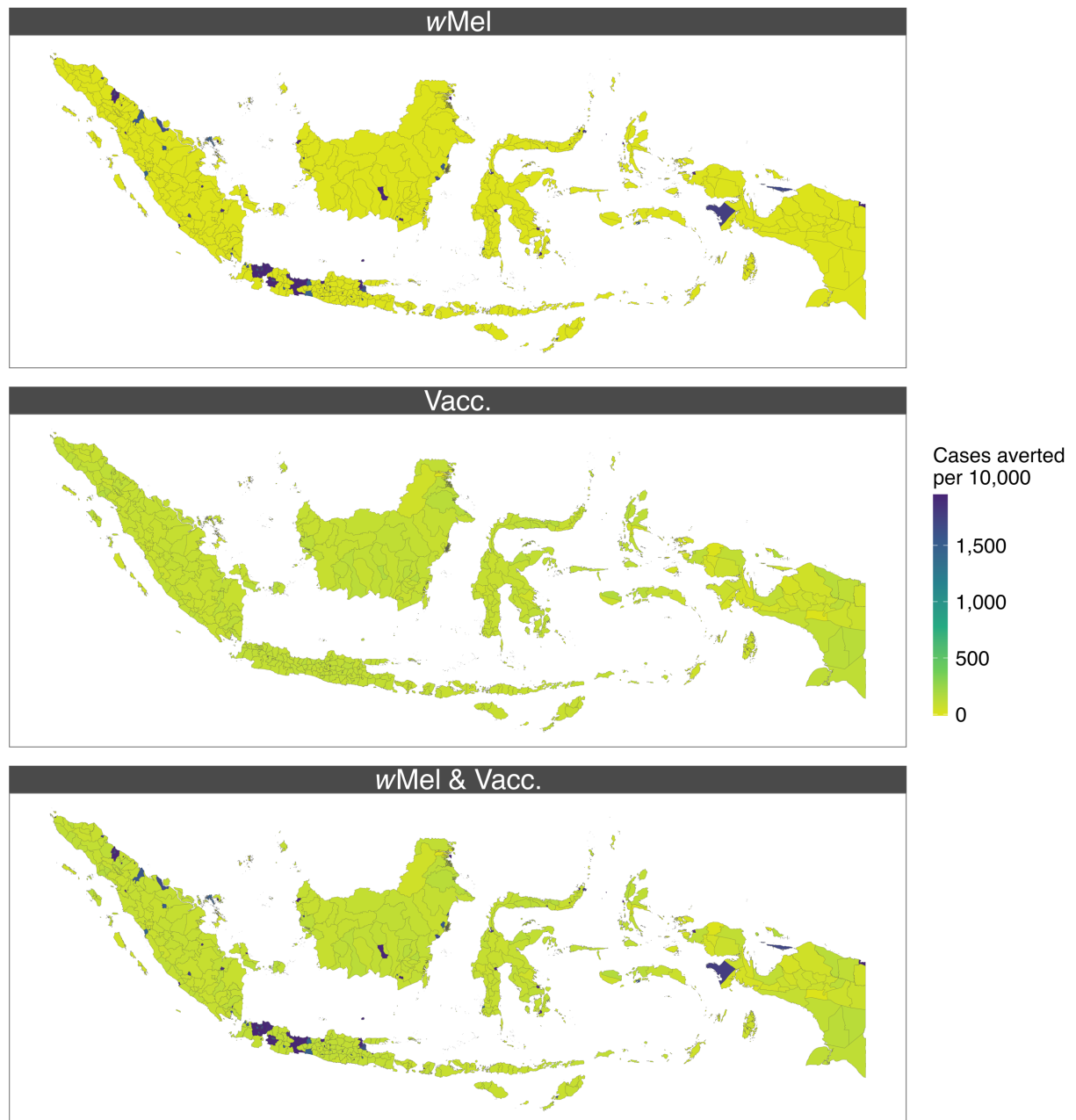
Public health impacts of wMel releases and vaccination over 15 years by municipality

Figure 4 | The population-level median cases averted per 10,000 individuals over 15 years by municipality in Indonesia over 500 stochastic simulations, comparing wMel only, vaccination only and wMel & vaccination. Yellow represents no cases averted. Higher numbers of cases averted are shown in dark green/blue.

Mean age dynamics

We estimated the national mean age of cases throughout each simulation, summarising with the median and 95% UI. The mean age of cases increased over the simulation period in the baseline scenario. Interventions further increased the mean age across the period, with the largest increases observed in the combined scenario.

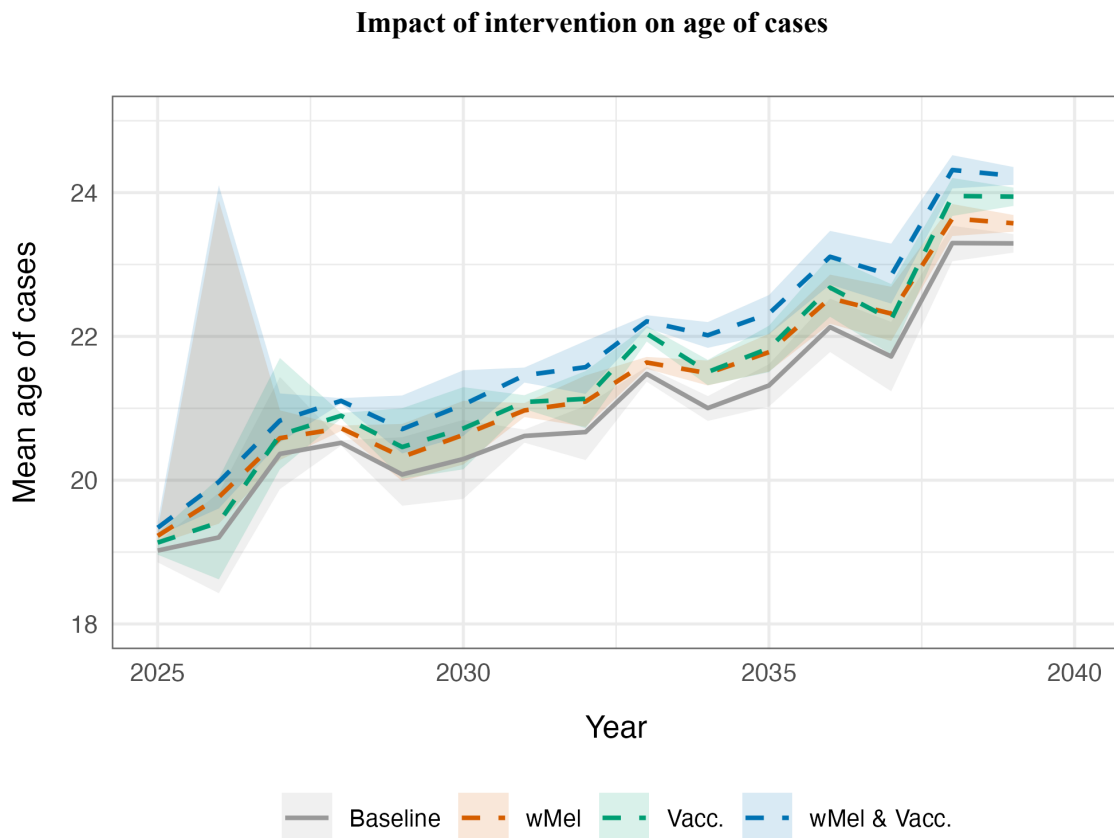


Figure 5 | The median of the national mean age of cases across 15 years, comparing no intervention (grey), wMel only (orange), vaccination only (green) and wMel & vaccination (blue). 95% credible intervals are shown.

Discussion

Population-level vs individual-level impact

The differential population- and individual-level impacts raises questions for policy makers regarding how to maximise public health impacts and optimally allocate resources. At the population-level, the scenarios including *Wolbachia* avert a much higher proportion of cases and hospitalisations than the vaccination-only scenario. This is because our selection criteria for *Wolbachia* release likely targeted municipalities which account for a high proportion of the burden, with O'Reilly et al. (2019) estimating that 90% of dengue cases occur in 15.3% of the land area. While the largest benefits are observed in the combined scenario, the relatively small difference between the burden averted in the wMel and combined intervention scenarios at the population-level (Figure 3) suggests that most of the impact is driven by *Wolbachia*. This implies that resources allocated to vaccination may be better invested into additional *Wolbachia* releases. Conversely, the individual-level results suggest that resources may be better allocated by expanding the cohorts targeted for vaccination. Future modelling studies could explore the alternative resource allocation scenarios to evaluate this.

The differences reflect the distinct mechanisms and targeted populations of the two interventions. *Wolbachia* reduces transmission across targeted municipalities, benefiting all individuals living in areas of successful introgression (Figure 4). Vaccination is assumed only to provide direct protection to vaccinated individuals, with no effect on transmission, due to its unclear efficacy against infection (El Hindi et al., 2025). Consequently, the benefits of vaccination are confined to vaccine recipients. The effects of wMel in the combined strategy protect individuals not eligible for vaccination, compensating for the assumed lack of indirect protection and potentially offering a more equitable solution than vaccination alone.

Assessing the individual-level impact is important because the benefits of vaccination accrue over time as more individuals are targeted. This pattern is clear in the individual-level trajectories, particularly for hospitalisations, where the vaccination and combined intervention scenarios diverge

more strongly in the later years compared to the baseline scenario than in the population-level results (Figure 2). While the population-level impact of vaccination after 15 years appears modest, the direct protective effect among vaccinated individuals is substantially higher. Additionally, the individual-level results highlight the larger impact of vaccination on severe dengue than milder cases, with a higher proportion of hospitalisations averted than cases (Figure 3).

Long-term effects

Simulations were carried out over a 15-year period. It would be useful to assess how the predicted intervention impact changes over a longer time frame. In particular, the population-level impact of vaccination would likely increase, as a higher proportion of the population benefits from the direct protective effects of vaccination.

Although epidemiological modelling cannot account for significant biological changes, some insight into the effects of interventions in the case of the emergence of resistance could be provided by tracking long-term changes in the proportion of susceptible individuals. Transmission-reducing interventions, such as *Wolbachia*, slow the accumulation of population immunity, while vaccination accelerates it. In the case of a largely naïve population due to long-term *Wolbachia* deployment, the emergence of a resistant DENV strain, capable of transmission in *Wolbachia*-carrying mosquitoes, could result in a very large outbreak. This could place considerable strain on healthcare systems.

Mean age

The increase in mean age of cases over the simulation period in the baseline scenario is driven by demographic aging, as the average age of an individual increased by 4.1 years throughout the period.

wMel and vaccination further shift the mean age upwards but through different mechanisms.

Wolbachia reduces transmission, delaying infections, while vaccination reduces disease risk in the younger vaccinated cohorts. Although the magnitude of these shifts are modest over the 15-year period, they may be more extreme over longer time periods. Severe outcomes are more common in older individuals (Yuan et al., 2022), meaning that shifts in the age distribution of cases could

partially offset the reductions in burden. Long-term studies modelling age-dependent hospitalisation would be useful to assess the impact of such demographic shifts.

Limitations

The model has several limitations which may influence the predicted intervention impact. First, the model does not track the infecting serotype but transmission dynamics are strongly influenced by the circulating serotype(s) because individuals can only be infected by each serotype once (Reich et al., 2013). TAK-003 has heterogeneous efficacy across serotypes, with lower efficacy against DENV-3 and DENV-4. As all four serotypes circulate in Indonesia (Sasmono et al., 2018), vaccination impacts would be underestimated when/where DENV-2 predominates circulation but overestimated when/where DENV-3 or DENV-4 predominate.

Second, we assumed immediate and successful introgression of *Wolbachia* across every municipality of release. This may result in the overestimation of the impact of *Wolbachia* in the first few years because *Wolbachia* requires a period of time for establishment and depends on local environmental conditions, such as temperature (Ross et al., 2017).

Third, we assumed no spatial coupling in transmission between municipalities. This may result in the underestimation of the impact of *Wolbachia* in later years as in reality the spread of wMel-carrying mosquitoes would likely lead to reductions in transmission in areas neighbouring targeted municipalities. Finally, FOI estimates varied little across municipalities. Improved serological data could help refine FOI estimates and targeting strategies.

Conclusion

Modelling enables potential intervention impacts to be assessed across large spatial scales and extended time frames that would be difficult to assess through field trials alone, helping inform public health decision-making. Our results suggest that combining *Wolbachia* and vaccination programs would likely be optimal, although the long-term effectiveness and best resource allocation between

the interventions remains uncertain. The long-term impacts may depend on factors which are currently difficult to predict, such as the evolutionary stability of the dengue-*Wolbachia*-mosquito interaction (Edenborough et al., 2021). Additionally, there is some uncertainty regarding the safety of TAK-003 in seronegative recipients against DENV-3 and DENV-4 (World Health Organization, 2024). Therefore, ongoing surveillance following the introduction of interventions will be essential for improving our understanding of intervention effectiveness, sustainability, and safety.

Materials and Methods

Data

We used published FOI estimates at the municipality (level 2 administrative unit) level (Cattarino et al., 2020). Estimates were available for 500 municipalities out of the 502 municipalities comprising Indonesia. The two municipalities without FOI estimates were not used.

We estimated reproduction numbers (R_0) from FOI estimates for each municipality, following the methodology outlined in Cattarino et al. (2020). We then derived the baseline transmission rate for each municipality as $\beta_n^0 = R_{0,n} \cdot \gamma$, where γ is the recovery rate. We constructed a demography dataset, stratified by year, age and municipality. We aggregated annual gridded population projections (Piontek and Geiger, 2024) into municipalities for 2020 to 2100. We stratified these projections into 5-year age groups (from 0 to 90+) using weights calculated from national population estimates and projections from the Wittgenstein Center (KC et al., 2024), after linear interpolation from 5-year to 1-year time steps. We split the population from each age group evenly between the constituent single-year age groups to produce a demography dataset stratified by year, age (0 to 90+) and municipality. We used data following the SSP2 intermediate climate change scenario from both sources.

Selection of wMel release sites

To assess the potential impacts of operationally feasible *Wolbachia* release, we selected 100 municipalities for wMel release. We selected municipalities based on a joint metric combining population-weighted density and average FOI estimates. Population-weighted density, pwd , is a measure of the population density experienced by an average individual living within a particular region (Craig, 1984). We calculated pwd for each municipality as the average population density across its constituent grid squares (30 arc resolution, approximately 1x1 km²), weighted by population size:

$$pwd = \frac{\sum (P_i \cdot d_i)}{\sum P_i}$$

where P_i is the population of subdivision i and d_i is the population density of subdivision i (Craig, 1984).

We used the WorldPop estimates for 2025 in our calculation (Bondarenko et al., 2025). Using min-max scaling, we scaled the $\log(pwd)$ and average FOI estimates (Cattarino et al., 2020) across municipalities. We calculated the joint metric for each municipality as the product of its scaled $\log(pwd)$ and FOI. We selected municipalities with the top 100 scores for *Wolbachia* deployment.

Transmission model

Model equations

We developed a stochastic transmission model using *odin2* (FitzJohn et al., 2021), with an S-I-C-R-V framework: the population is divided into susceptible ($S_{a,n}$), infected ($I_{a,n,k}$), cross-protected ($C_{a,n,k}$), recovered ($R_{a,n,k}$) and vaccinated ($V_{a,n,k}$) compartments (Figure S6). The model is stratified by age (using single age groups from 0 to 90+) and municipality. The number of previous infections is tracked but infecting serotypes are not tracked. Individuals can experience up to four infections, after which they obtain lifelong immunity against further infection.

We modelled dengue transmission independently within each municipality, with no spatial coupling in transmission between the municipalities. The model assumes homogeneous mixing within municipalities.

We implemented a discrete time modelling framework following the equations below. For non-ageing days, the daily transitions are:

$$\begin{aligned}
 S_{a,n}(t+1) &= S_{a,n}(t) + \delta_{a,1}B_n(t) - n_{a,n}^{SI}(t) - v_{a,n}(t) \cdot S_{a,n}(t) + n_{a,n,1}^{wn}(t) \\
 I_{a,n,k}(t+1) &= \begin{cases} I_{a,n,1}(t) + n_{a,n}^{SI}(t) + n_{a,n,1}^{VI}(t) - n_{a,n,1}^{IC}(t), & k = 1 \\ I_{a,n,k}(t) + n_{a,n,k-1}^{RI}(t) + n_{a,n,k}^{VI}(t) - n_{a,n,k}^{IC}(t), & k = 2,3,4 \end{cases} \\
 C_{a,n,k}(t+1) &= C_{a,n,k}(t) + n_{a,n,k}^{IC}(t) - n_{a,n,k}^{CR}(t), \quad k = 1,2,3,4 \\
 R_{a,n,k}(t+1) &= R_{a,n,k}(t) + n_{a,n,k}^{CR}(t) - n_{a,n,k}^{RI}(t) - v_{a,n}(t) \cdot R_{a,n,k}(t) + n_{a,n,k+1}^{wn}(t), \quad k = 1,2,3,4 \\
 V_{a,n,k}(t+1) &= \begin{cases} V_{a,n,1}(t) + v_{a,n}(t) \cdot S_{a,n}(t) - n_{a,n,1}^{VI}(t) - n_{a,n,1}^{wn}(t), & k = 1 \\ V_{a,n,k}(t) + v_{a,n}(t) \cdot R_{a,n,k-1}(t) - n_{a,n,k}^{VI}(t) - n_{a,n,k}^{wn}(t), & k = 2,3,4,5 \end{cases}
 \end{aligned}$$

$\delta_{x,y}$ is the Kronecker delta function which is equal to 1 when $x = y$ and otherwise 0. Compartment S is stratified by age, a , and region, n . Compartments I , C and R are additionally stratified by the number of previous infections, k which can take values 1 to 4. Compartment V is stratified by a , n and k , where $k - 1$ represents the number of previous infections prior to vaccination. $B(t)$ represents the number of births. $n_{a,n,1}^{wn}(t)$ represents the number of individuals who transition from compartment $V_{a,n,1}$ to $S_{a,n}$, due to vaccine immunity waning. $n_{a,n,k+1}^{wn}(t)$ represents the number of individuals who transition from $V_{a,n,k+1}$ to $R_{a,n,k}$. $n_{a,n,k}^{RI}(t)$ represents the number of individuals who transition from $R_{a,n,k}$ to $I_{a,n,k+1}$. Otherwise, $n_{a,n,k}^{XY}(t)$ represents the number of individuals who transition from $X_{a,n,k}$ to $Y_{a,n,k}$. $v_{a,n}(t)$ is the proportion vaccinated. See Tables S3, S4 and S5 for more detail.

The number of individuals vaccinated is determined by $v_{a,n}(t) \cdot S_{a,n}(t)$ and $v_{a,n}(t) \cdot R_{a,n,k-1}(t)$.

Other transitions between compartments are modelled as draws from binomial distributions (Tables S1 and S5):

$$\begin{aligned}
 n_{a,n}^{SI}(t) &\sim \text{Binomial}\left(S_{a,n}(t), p_{n,1}^{inf}(t)\right) \\
 n_{a,n,k}^{IC}(t) &\sim \text{Binomial}\left(I_{a,n,k}(t), p^\gamma\right), \quad k = 1, 2, 3, 4 \\
 n_{a,n,k}^{CR}(t) &\sim \text{Binomial}\left(C_{a,n,k}(t), p^\nu\right), \quad k = 1, 2, 3, 4 \\
 n_{a,n,k}^{RI}(t) &\sim \text{Binomial}\left(R_{a,n,k}(t), p_{n,k+1}^{inf}(t)\right), \quad k = 1, 2, 3 \\
 n_{a,n,k}^{wn}(t) &\sim \text{Binomial}\left(V_{a,n,k}(t), p^{wn}\right), \quad k = 1, 2, 3, 4, 5 \\
 n_{a,n,1}^{VI}(t) &\sim \text{Binomial}\left((V_{a,n,1}(t) - n_{a,n,1}^{wn}(t)), p_{n,1}^{inf}(t)(1 - \varepsilon_1^{inf})\right) \\
 n_{a,n,k}^{VI}(t) &\sim \text{Binomial}\left((V_{a,n,k}(t) - n_{a,n,k}^{wn}(t)), p_{n,k}^{inf}(t)(1 - \varepsilon_2^{inf})\right), \quad k = 2, 3 \\
 n_{a,n,k}^{VI}(t) &= 0, \quad k = 4, 5
 \end{aligned}$$

with transition probabilities (Tables S1 and S6):

$$\begin{aligned}
 p_{n,k}^{inf}(t) &= 1 - \exp(-\lambda_n(t) \cdot \frac{5-k}{4} \cdot dt) \\
 p^\gamma &= 1 - \exp(-\gamma \cdot dt) \\
 p^\nu &= 1 - \exp(-\nu \cdot dt) \\
 p^{wn} &= 1 - \exp\left(-\frac{dt}{\omega \cdot 365}\right)
 \end{aligned}$$

where $\lambda_n(t)$ is the FOI at time t , dt is the model time step, γ is the rate of recovery ($days^{-1}$), ν is the rate of loss of cross-protection ($days^{-1}$), ω is the mean duration of vaccine induced protection (years).

$\lambda_n(t)$, is calculated as:

$$\lambda_n(t) = \beta_n(t) \cdot \frac{\sum_a \sum_{k=1}^4 I_{a,n,k}(t)}{\sum_a N_{a,n}(t)}$$

where $N_{a,n}(t)$ is the total population and $\beta_n(t)$ is the transmission rate, calculated as:

$$\beta_n(t) = \beta_n^0 \cdot \left(1 + A \cos\left(\frac{2\pi t}{365} - \phi\right)\right) \cdot \begin{cases} (1 - \omega_{wol}), & \text{if Wolbachia is deployed in municipality } n \\ 1, & \text{otherwise} \end{cases}$$

where β_n^0 is the baseline transmission rate, A is the amplitude of seasonal forcing, ϕ is the phase of seasonal forcing and ω_{wol} is the proportional reduction in the transmission rate due to wMel (Table S1).

We modelled cases and hospitalisations as deterministic functions of the number of new infections each day. We assumed that the probability of experiencing symptoms once infected varies by the number of previous infections, with secondary infection having the highest probability of symptoms (Cracknell Daniels, 2024). We assumed that the probability of hospitalisation given a symptomatic case is 0.09 (Ferguson et al., 2016). For unvaccinated individuals, the daily number of symptomatic cases and hospitalisations in age group a in region n are:

$$cases_{a,n}(t) = n_{a,n}^{SI} \sigma_1 + n_{a,n,1}^{RI} \sigma_2 + n_{a,n,2}^{RI} \sigma_3 + n_{a,n,3}^{RI} \sigma_4$$

$$hosp_{a,n}(t) = n_{a,n}^{SI} \sigma_1 h + n_{a,n,1}^{RI} \sigma_2 h + n_{a,n,2}^{RI} \sigma_3 h + n_{a,n,3}^{RI} \sigma_4 h$$

where σ_k is the probability of a symptomatic case given the k^{th} infection and h is the probability of hospitalisation given a symptomatic case (Table S1).

For vaccinated individuals who experience breakthrough infections, vaccine efficacy against symptomatic disease and hospitalisation (see Figure S7 for derivation) is applied (Table S1):

$$\begin{aligned} cases_{a,n}^V(t) &= n_{a,n,1}^{VI} \sigma_1 (1 - \varepsilon_1^{symp|inf}) + n_{a,n,2}^{VI} \sigma_2 (1 - \varepsilon_2^{symp|inf}) + n_{a,n,3}^{VI} \sigma_3 (1 - \varepsilon_2^{symp|inf}) \\ hosp_{a,n}^V(t) &= n_{a,n,1}^{VI} \sigma_1 h (1 - \varepsilon_1^{hosp|inf}) + n_{a,n,2}^{VI} \sigma_2 h (1 - \varepsilon_2^{hosp|inf}) + n_{a,n,3}^{VI} \sigma_3 h (1 - \varepsilon_2^{hosp|inf}) \end{aligned}$$

where $\varepsilon_s^{symp|inf}$ is the conditional vaccine efficacy against symptomatic disease given that infection has occurred

$$\varepsilon_s^{symp|inf} = \frac{\varepsilon_s^{symp} - \varepsilon_s^{inf}}{1 - \varepsilon_s^{inf}}, \quad s = 1, 2$$

and $\varepsilon_s^{hosp|inf}$ is the conditional vaccine efficacy against hospitalisation given that infection has occurred

$$\varepsilon_s^{hosp|inf} = \frac{\varepsilon_s^{hosp} - \varepsilon_s^{inf}}{1 - \varepsilon_s^{inf}}, \quad s = 1, 2.$$

The total daily symptomatic cases and hospitalisations is calculated as:

$$\begin{aligned} cases_{a,n}^{total}(t) &= cases_{a,n}(t) + cases_{a,n}^V(t) \\ hosp_{a,n}^{total}(t) &= hosp_{a,n}(t) + hosp_{a,n}^V(t) \end{aligned}$$

The annual number of births for each region was extracted from the demographic data and spread out to occur evenly throughout the year. We used a cohort-based model of ageing where individuals move up an age class once per year. On the ageing day, each age class is scaled to match the population size for that region according to age-stratified demographic projections. Each year, we introduce annual seeding of infection in each municipality proportional to population size (summing to 100 nationally) on a randomly drawn day ($Uniform(2, 365)$) in a randomly drawn age group ($Uniform(10, 45)$).

Vaccination

Vaccination occurs once per year on the day after ageing. A proportion of individuals in susceptible or recovered compartments in the eligible age cohort move to the vaccinated compartments in line with the assumed vaccine coverage. We modelled a 3-year vaccination catch-up regime for individuals aged 10-16 years, followed by yearly vaccination of 10-year-olds, with a vaccine coverage of 60%.

We used estimates from the phase III trial data of TAK-003 after 3 years for the serostatus-specific vaccine efficacy against symptomatic dengue and hospitalisation (Rivera et al., 2022). We assumed that vaccination provides no protection against infection.

wMel releases

We modelled the impact of wMel release as a reduction on the transmission rate, β_n , producing a proportional reduction in FOI, λ_n . We assumed that wMel reduced the transmission rate by 50% (Cavany et al., 2023), with immediate 100% introgression.

Counterfactual comparisons

We simulated 500 stochastic simulations for the following scenarios:

1. Baseline: No intervention scenario
2. *Wolbachia*: wMel releases in 100 municipalities, assuming 50% reduction in the transmission rate.
3. Vaccination: National vaccination program with 60% coverage. 3-year catch up campaign targeting 10–16-year-olds followed by routine vaccination of 10-year-olds.
4. Combined *Wolbachia* and vaccination scenario

See Table S1 for the full set of parameter values.

For the non-vaccine scenarios, individuals were ‘vaccinated’ with the same coverage but with efficacies of zero (Table S1). This enabled a comparison of the individual-level impact of vaccination within the same cohort of individuals, following Cracknell Daniels et al. (2025).

Individual-level impact refers to the impact within the vaccinated cohort only (or the equivalent population in the scenarios without vaccination). Population-level impact refers to the impact across the entire population, including both vaccinated and non-vaccinated individuals.

We summarised the number of cases and hospitalisations with the median and 95% UI, for each scenario, across the 500 simulations at both national and municipality-based levels and at both population- and individual-levels. We calculated the number and proportion of cases and hospitalisations averted compared to the baseline scenario for each run at both national and municipality-based levels and summarised with the median and 95% UI. To produce the maps of intervention impact, we calculated the population-level median cumulative cases and hospitalisations averted per 10,000 individuals after 15 years by municipality:

$$averted\ per\ 10,000 = \frac{median\ cumulative\ averted}{municipality\ population\ in\ 2025} \times 10,000$$

We calculated the national cases-weighted mean age occurring in each year for each simulation, where

$$mean\ age_{particle} = \frac{\sum_a age_a \times cases_a}{\sum_a cases_a},$$

and summarised with the median and 95% UI.

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